

# Sentiment analysis for Mental Health Status Classification

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**Abstract**—Sentiment analysis, a key task in natural language processing (NLP), aims to determine the emotional tone behind a body of text. This study presents a comparative approach using both classical machine learning and transformer-based deep learning models. The dataset, composed of text and sentiment labels, is preprocessed using traditional NLP techniques, vectorized using TF-IDF for classical models, and tokenized with BERT for deep learning. We implement and evaluate XGBoost and BERT-based models, analyzing performance through accuracy, confusion matrices, and classification reports. Our findings highlight the strengths and trade-offs of both approaches for sentiment classification tasks.

## I. PROBLEM STATEMENT AND OBJECTIVES

### A. Problem Statement

With the rapid growth of user-generated text on digital platforms, sentiment analysis has become essential for extracting insights from opinions and reviews. Traditional machine learning models, while efficient, often lack contextual understanding. On the other hand, transformer-based models like BERT offer deeper language comprehension but are resource-intensive.

This research aims to bridge this gap by evaluating and comparing the effectiveness of a classical model (XGBoost) and a transformer-based model (BERT) for sentiment classification. The goal is to identify a balanced approach that optimizes both performance and computational efficiency.

### B. Research Objectives

The primary aim of this research is to develop and evaluate sentiment analysis models using both classical machine learning algorithms and transformer-based deep learning architectures. This dual approach allows for a comprehensive comparison of model performance, scalability, and interpretability in handling textual sentiment classification tasks.

## II. METHODOLOGY

### A. Data Collection

The dataset used in this study, `Combined Data.csv`, comprises textual entries accompanied by sentiment labels such as *stress*, *anxiety*, and *normal*. The data was read using the `pandas` library in Python.

### B. Data Preprocessing

Preprocessing was a crucial step to ensure clean and standardized input for modeling. The following operations were performed:

- **URL and HTML Removal:** Regex patterns were used to remove URLs and HTML tags.
- **Handle Removal:** Mentions such as @usernames were stripped from the text.
- **Case Normalization:** All text was converted to lower-case.
- **Punctuation Removal:** Non-alphabetic characters and punctuation were removed.
- **Stopword Removal:** Common English stopwords were eliminated using the NLTK library.

### C. Downsampling

To address the issue of class imbalance in the original dataset, a downsampling approach was applied prior to model training. The criterion for downsampling was based on the frequency of the least represented class in the dataset. All other classes were reduced to match this count, ensuring a uniform class distribution. This preprocessing step was critical to prevent bias toward majority classes and to improve the classifier's ability to generalize across all mental health categories.

### D. Feature Engineering

1) **TF-IDF Vectorization:** For the classical machine learning pipeline, text was converted into numerical vectors using Term Frequency–Inverse Document Frequency (TF-IDF). This technique captures word relevance within documents while penalizing frequent terms that occur across the corpus.

2) **BERT Tokenization:** For the deep learning model, textual data was tokenized using the BERT tokenizer from the HuggingFace Transformers library. Tokens were converted into input IDs, attention masks, and segment embeddings compatible with BERT input formats.

### E. Label Encoding

Sentiment labels were encoded numerically using the Scikit-learn `LabelEncoder`. This encoding was necessary to train classification models effectively.

### F. Model Development

1) *XGBoost Classifier*: XGBoost, a scalable and efficient gradient boosting framework, was used as the primary classical model. The classifier was trained on TF-IDF features and optimized using default and tuned hyperparameters.

2) *BERT Model*: The BERT model (DistilBERT, uncased) was used to generate contextualized word embeddings. These embeddings were passed through a dense classification head for final sentiment prediction. Training leveraged the transformers and torch libraries.

### G. Model Training and Evaluation

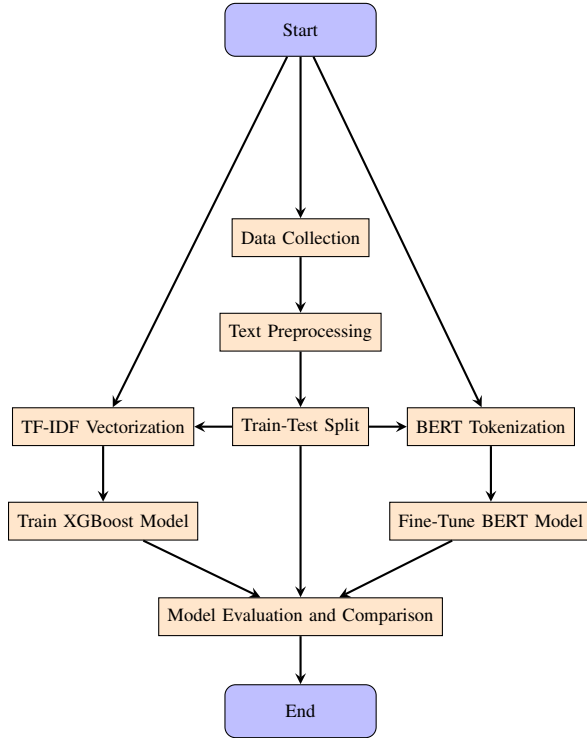
The dataset was split into training and testing sets using an 80/20 ratio. The following metrics were used for model evaluation:

- **Accuracy Score**
- **Confusion Matrix**
- **Precision, Recall, and F1-score**

## III. PROJECT FLOWCHART

### A. Flowchart of Sentiment Analysis Process

The following flowchart outlines the end-to-end process followed in the sentiment analysis project, incorporating both classical machine learning and transformer-based deep learning approaches:



## IV. SOFTWARE REQUIREMENT

The implementation framework leverages a diverse set of specialized libraries, each optimized for distinct stages of the machine learning pipeline. As outlined in Table 1, libraries such as pandas and numpy facilitate efficient data manipulation, while matplotlib and seaborn support advanced

data visualization. Traditional machine learning algorithms, including gradient boosting, are implemented using xgboost. For transformer-based architectures like BERT, we employ the transformers and datasets libraries to handle model loading and dataset management. Finally, model training and evaluation workflows are orchestrated with scikit-learn and PyTorch to ensure scalable and reproducible experimentation.

TABLE I  
REQUIRED TOOLS BY FUNCTIONALITY

| Functionality                   | Tools used             |
|---------------------------------|------------------------|
| IDE                             | Google Colab           |
| Scripting                       | Python3                |
| Data manipulation               | Pandas, Numpy          |
| Plotting                        | Matplotlib, Seaborn    |
| Traditional ML models (XGBoost) | Xgboost                |
| Transformers/BERT model         | Transformers, Datasets |
| Model training and evaluation   | Scikit-learn, Torch    |

## V. CHALLENGES

- **Session Time Limits**: Free trial version of Google colab have limited GPU/TPU session durations per day.
- **GPU/TPU Compatibility**: CUDA version mismatches caused library import errors.

## VI. RESULTS

This section presents the performance evaluation of the sentiment classification system applied to a balanced dataset derived through downsampling. The model was tested using various metrics, including accuracy, precision, recall, and F1-score.

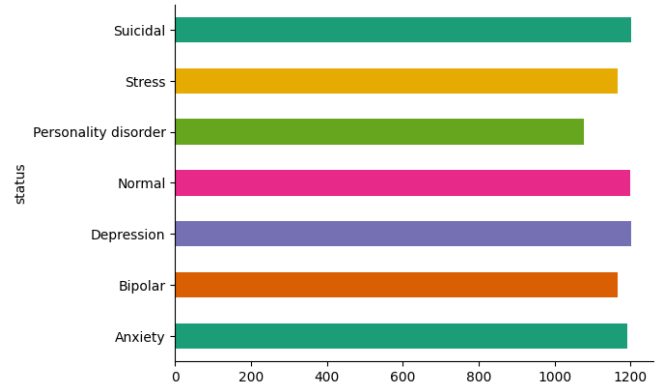


Fig. 1. Number of data points for each class after downsampling

### A. Evaluation metrics of XGboost

The XGBoost classifier, trained on TF-IDF features extracted from the cleaned and downsampled text data, achieved an overall accuracy of **71.72%**. Table II provides a detailed breakdown of precision, recall, and F1-score for each mental health category.

TABLE II  
PERFORMANCE METRICS OF XGBOOST MODEL

| Class                   | Precision     | Recall | F1-Score | Support |
|-------------------------|---------------|--------|----------|---------|
| Anxiety                 | 0.75          | 0.74   | 0.74     | 242     |
| Bipolar                 | 0.92          | 0.78   | 0.84     | 249     |
| Depression              | 0.60          | 0.54   | 0.57     | 223     |
| Normal                  | 0.74          | 0.90   | 0.81     | 268     |
| Personality Disorder    | 0.78          | 0.73   | 0.75     | 215     |
| Stress                  | 0.64          | 0.62   | 0.63     | 222     |
| Suicidal                | 0.60          | 0.67   | 0.63     | 222     |
| <b>Overall Accuracy</b> | <b>71.72%</b> |        |          |         |
| <b>Macro Average</b>    | 0.72          | 0.71   | 0.71     | 1641    |
| <b>Weighted Average</b> | 0.72          | 0.72   | 0.72     | 1641    |

1) *Analysis of Model Performance:* The classifier showed strong performance in identifying Bipolar and Normal classes, achieving F1-scores of 0.84 and 0.81 respectively. These categories likely exhibit more distinct lexical patterns, making them easier to identify.

Moderate performance was observed in the Anxiety, Personality Disorder, and Stress categories, with F1-scores ranging between 0.63 and 0.75. These results suggest a fair degree of class distinguishability after downsampling.

Relatively lower scores were recorded for Depression and Suicidal classes, with F1-scores of 0.57 and 0.63 respectively. The overlapping vocabulary between these emotional states and others may have contributed to misclassifications.

2) *Interpretation:* The model's overall macro-average F1-score of **0.71** indicates balanced performance across categories. These results validate the effectiveness of TF-IDF-based features and downsampling for handling multi-class mental health classification. However, further improvements may be achieved through ensemble methods or deep contextual models such as BERT fine-tuning.

#### B. Evaluation metrics of BERT model

This section presents the evaluation outcomes of the BERT-based sentiment classification model applied to the downsampled mental health dataset. Performance metrics including precision, recall, F1-score, and support for each class were recorded and are summarized in Table III.

TABLE III  
PERFORMANCE METRICS OF BERT MODEL

| Class                   | Precision     | Recall | F1-Score | Support |
|-------------------------|---------------|--------|----------|---------|
| Anxiety                 | 0.86          | 0.80   | 0.83     | 215     |
| Bipolar                 | 0.89          | 0.87   | 0.88     | 215     |
| Depression              | 0.72          | 0.63   | 0.67     | 215     |
| Normal                  | 0.91          | 0.90   | 0.90     | 216     |
| Personality Disorder    | 0.79          | 0.88   | 0.84     | 216     |
| Stress                  | 0.75          | 0.78   | 0.76     | 215     |
| Suicidal                | 0.71          | 0.76   | 0.74     | 216     |
| <b>Overall Accuracy</b> | <b>80.37%</b> |        |          |         |
| <b>Macro Avg</b>        | 0.80          | 0.80   | 0.80     | 1508    |
| <b>Weighted Avg</b>     | 0.80          | 0.80   | 0.80     | 1508    |

1) *Classification Performance:* The BERT model achieved an overall accuracy of **80%** across seven mental health classes. The highest precision was observed for the *Normal* class (0.91), followed closely by *Bipolar* (0.89) and *Anxiety* (0.86). The *Depression* and *Suicidal* classes had lower performance metrics, suggesting overlapping features or less distinctive language usage in these categories.

In terms of recall, *Personality Disorder* exhibited the strongest retrieval performance (0.88), indicating the model's effectiveness in correctly identifying this class. The macro-averaged F1-score was 0.80, indicating balanced performance across all classes despite downsampling.



Fig. 2. Losses and accuracy of BERT over the training epochs

## VII. CONCLUSION

This study evaluated the performance of two distinct machine learning approaches—XGBoost with TF-IDF vectorization and BERT with pre-trained tokenization—on the task of multi-class classification for mental health status based on sentiment analysis. The results demonstrate that the BERT model significantly outperforms the traditional XGBoost classifier across all evaluation metrics.

The BERT model achieved an overall accuracy of 80.37%, a macro-average F1-score of 0.80, and a weighted-average F1-score of 0.80, indicating consistent and robust performance across all mental health categories. In contrast, the XGBoost model attained an overall accuracy of 71.72%, with a macro-average F1-score of 0.71 and a weighted-average F1-score of 0.72. Notably, BERT showed substantial improvements in identifying critical classes such as Depression (F1-score: 0.67 vs. 0.57) and Suicidal tendencies (F1-score: 0.74 vs. 0.63), which are of high clinical relevance.

## REFERENCES

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## APPENDIX

**Weblink to dataset:-** <https://www.kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health?select=Combined+Data.csv>.

TABLE IV  
SAMPLE STATEMENTS AND ASSOCIATED STATUSES IN DATASET

| Unnamed: 0 | statement                                          | status               |
|------------|----------------------------------------------------|----------------------|
| 45700      | bought a new phone when I didn't need one don't... | Bipolar              |
| 51490      | PMS and AvPD I have been feeling okay for quit...  | Personality disorder |
| 51142      | gender rules & being in a relationship as ...      | Personality disorder |
| 36155      | I don't hate the player I hate the game, Jes...    | Suicidal             |
| 31371      | it's a lot better than paying someone \$60.        | Normal               |